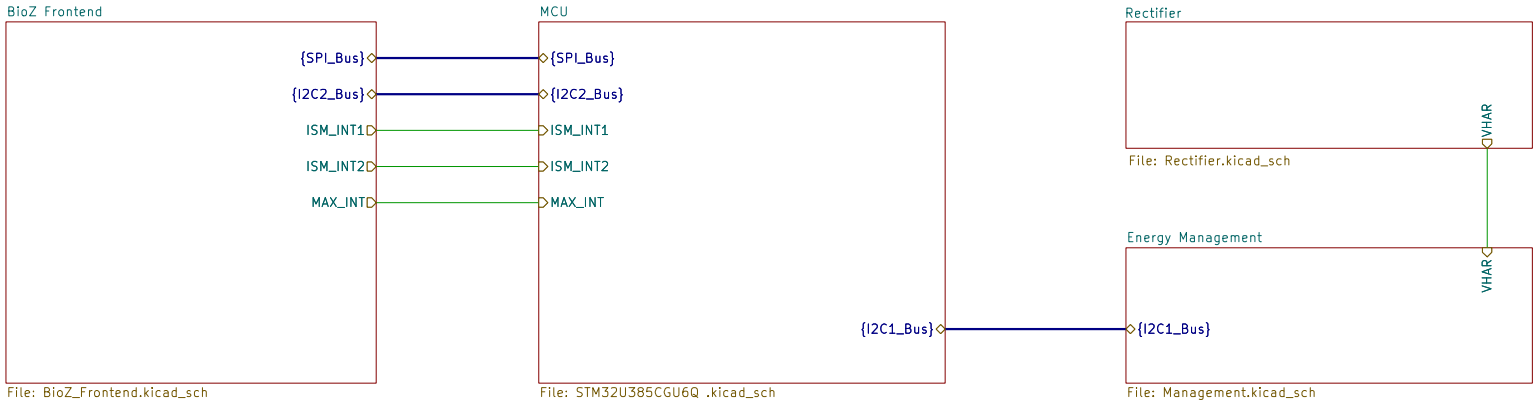
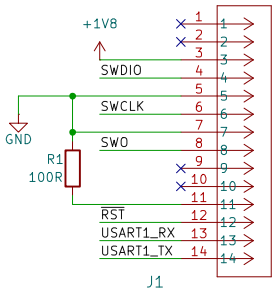
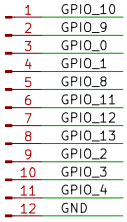


ETH zürich Lab: PBL Info: lkoehli@ethz.ch ProjID:	*Project Title:* *Design of an EQS Energy–Harvesting Frontend* **Root**
Date: Rev: KiCad E.D.A. 10.0.0	Drawn By: Lucas Köhli Checked By: Lukas Schulthess License:
File: BSc_Thesis.kicad_sch Sheet: /	Format: A4
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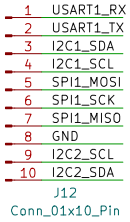
ST_Link Connection



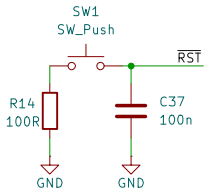
GPIO Pins



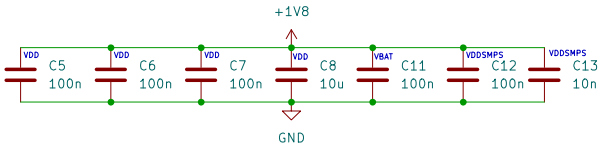
Communication Signals



Reset Switch

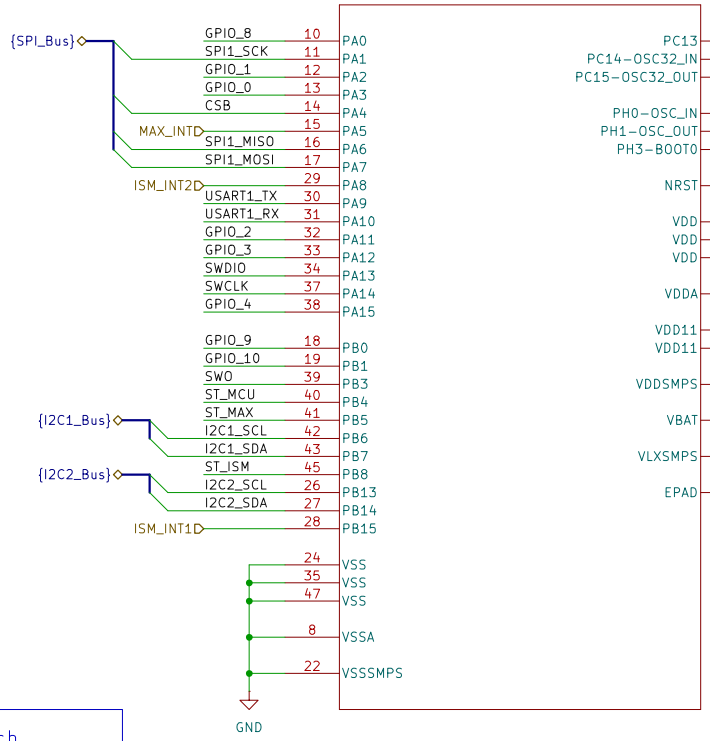


Decoupling Capacitors (VDD inputs)

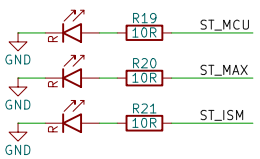


Microcontroller Unit

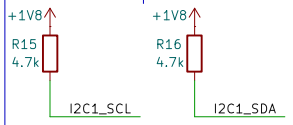
U1
STM32U385CGU6Q



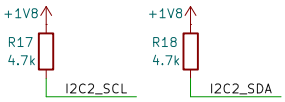
Status LED's



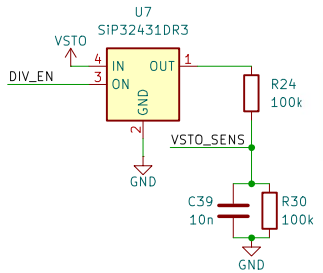
I2C1 Pull-up Resistors



I2C2 Pull-up Resistors

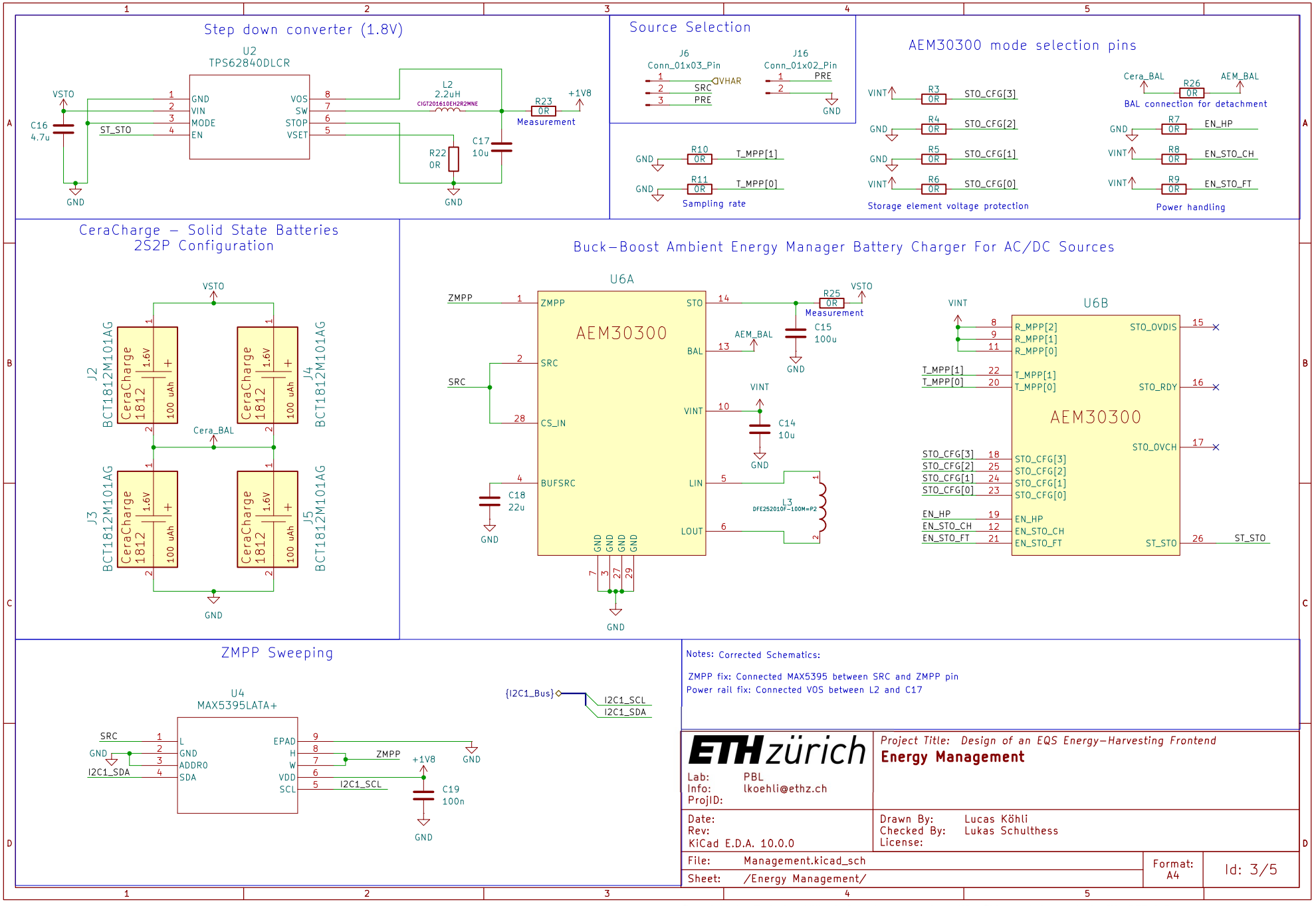


Storage Voltage Monitor (VSTO)

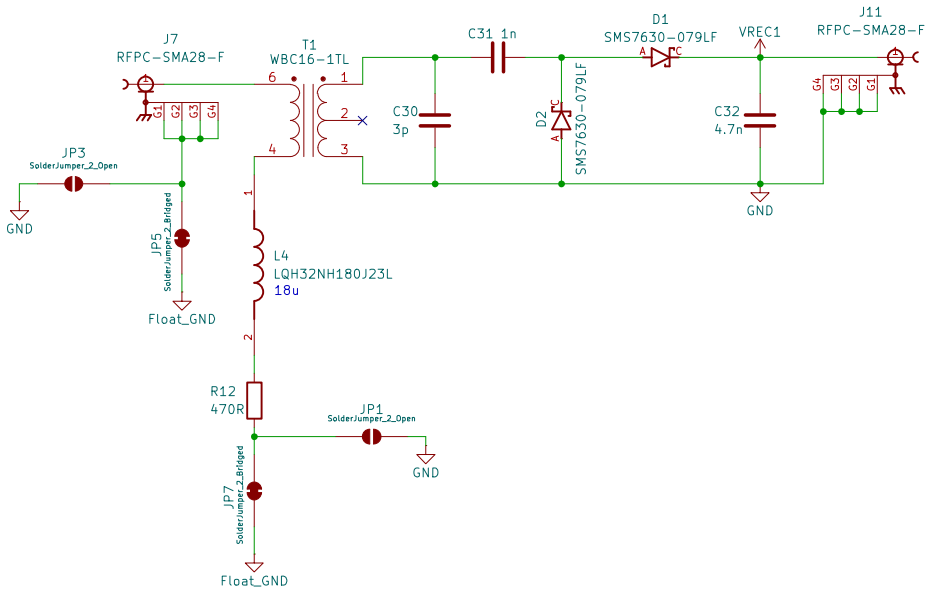


Notes:

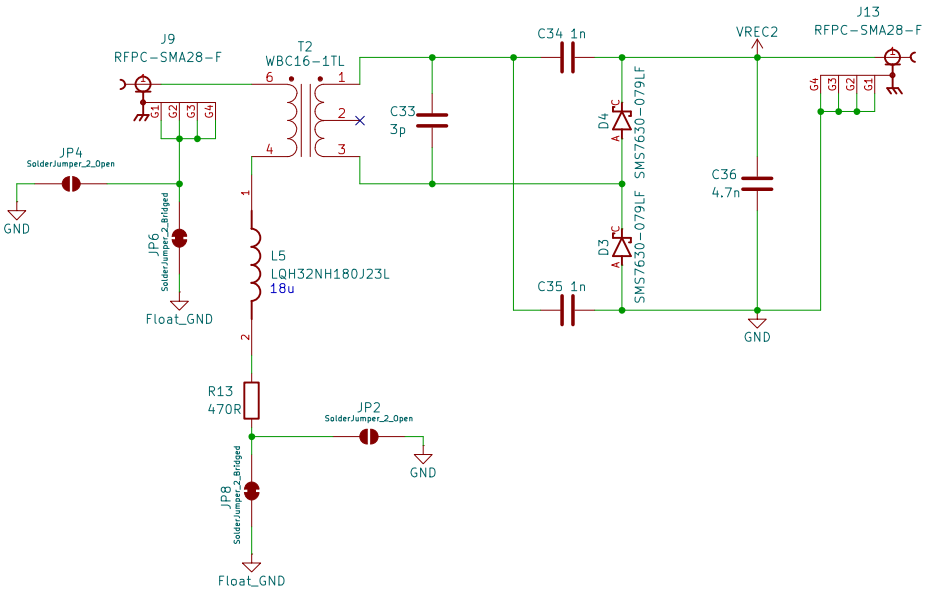
ETH zürich		Project Title: Design of an EQS Energy-Harvesting Frontend	
Lab: PBL Info: lkoehli@ethz.ch ProjID:		MCU	
Date: Rev: KiCad E.D.A. 10.0.0		Drawn By: Lucas Köhli Checked By: Lukas Schulthess License:	
File:	STM32U385CGU6Q .kicad_sch	Format: A4	Id: 2/5
Sheet:	/MCU/		


Step down converter (1.8V)
U2
TPS62840DLCR
VSTO
C16
4.7u
ST_STO
GND
VIN
MODE
EN
VOS
SW
STOP
VSET
L2
2.2uH
CIGT201610EH2R2MNE
R23
0R
+1V8
Measurement
R22
0R
C17
10u
Source Selection
J6
Conn_01x03_Pin
VHAR
SRC
PRE
J16
Conn_01x02_Pin
PRE
R10
T_MPP[1]
R11
T_MPP[0]
Sampling rate
AEM30300 mode selection pins
VINT
R3
STO_CFG[3]
R4
STO_CFG[2]
R5
STO_CFG[1]
R6
STO_CFG[0]
Storage element voltage protection
Cera_BAL
R26
AEM_BAL
BAL connection for detachment
R7
EN_HP
R8
EN_STO_CH
R9
EN_STO_FT
Power handling
CeraCharge – Solid State Batteries
2S2P Configuration
J2
BCT1812M101AG
J3
J4
J5
CeraCharge
1812
1.6V
100 uAh
Cera_BAL
Buck–Boost Ambient Energy Manager Battery Charger For AC/DC Sources
U6A
AEM30300
ZMPP
SRC
CS_IN
BUFSRC
STO
BAL
VINT
LIN
LOUT
C18
22u
C15
100u
C14
10u
L3
DFE252010F-100M=P2
U6B
R_MPP[2]
R_MPP[1]
R_MPP[0]
T_MPP[1]
T_MPP[0]
STO_OVDIS
STO_RDY
STO_OVCH
STO_CFG[3]
STO_CFG[2]
STO_CFG[1]
STO_CFG[0]
EN_HP
EN_STO_CH
EN_STO_FT
ST_STO
ZMPP Sweeping
U4
MAX5395LATA+
L
ADDR0
SDA
SCL
EPAD
H
W
VDD
{I2C1_Bus}
I2C1_SCL
I2C1_SDA
C19
100n
Notes: Corrected Schematics:
ZMPP fix: Connected MAX5395 between SRC and ZMPP pin
Power rail fix: Connected VOS between L2 and C17
ETH zürich
Lab: PBL
Info: lkoehli@ethz.ch
ProjID:
Project Title: Design of an EQS Energy–Harvesting Frontend
Energy Management
Date:
Rev:
KiCad E.D.A. 10.0.0
Drawn By: Lucas Köhli
Checked By: Lukas Schulthess
License:
File: Management.kicad_sch
Sheet: /Energy Management/
Format: A4
Id: 3/5

1. Topology: Half-Wave Voltage Doubler



2. Topology: Full-Wave Voltage Doubler



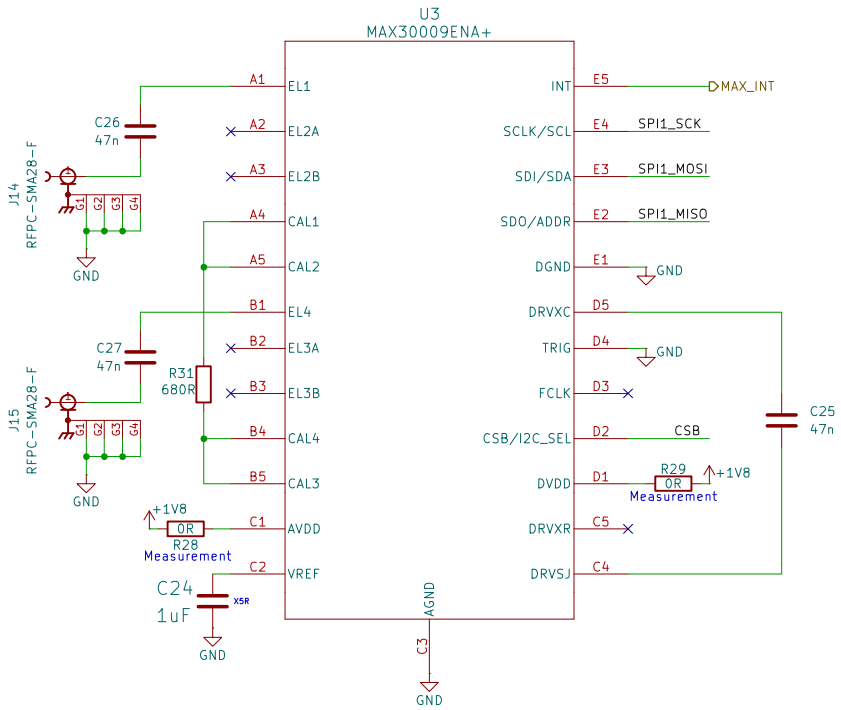
Source Selection



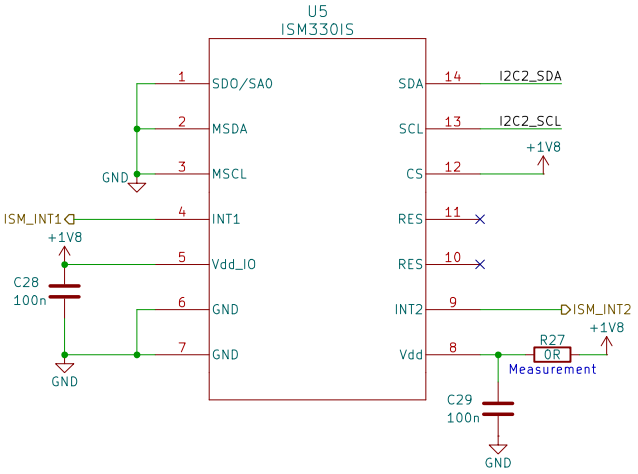
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ETH zürich		Project Title: Design of an EQS Energy-Harvesting Frontend **Rectifier**	
Lab: PBL Info: lkoehli@ethz.ch ProjID:			
Date: 08.04.2026 Rev: KiCad E.D.A. 10.0.0		Drawn By: Lucas Köhli Checked By: Lukas Schulthess License:	
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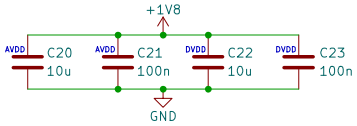
Low-Power, High-Performance Bioimpedance Analog Front-End
Bipolar (2-Electrode) Configuration



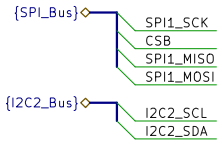
6-axis IMU:
Always-on 3-axis accelerometer and 3-axis gyroscope with ISPU



Decoupling Capacitors
(VDD Inputs – Max Chip)



SPI/I2C2 Bus



Notes:

ETH zürich Lab: PBL Info: lkoehli@ethz.ch ProjID:	*Project Title: Design of an EQS Energy-Harvesting Frontend* **BioZ Frontend**
Date: Rev: KiCad E.D.A. 10.0.0	Drawn By: Lucas Köhli Checked By: Lukas Schulthess License:
File: BioZ_Frontend.kicad_sch	Format: A4
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